

Restricted-Product Adelic Duality and the Loeb-Rosser Phantom

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Abstract

Pass 67 checks finite conductor shadows of the restricted product

$$\mathbb{A}_f = \prod_p' (\mathbb{Q}_p, \mathbb{Z}_p).$$

The finite local pieces have the expected self-duality, but the phantom $\widehat{\mathbb{Z}}/\mathbb{Z}$ is not visible at any fixed finite conductor level. It must be treated as a derived/pro quotient.

Main Result

For a prime p and conductor $k \geq 1$, use

$$p^{-k}\mathbb{Z}_p/p^k\mathbb{Z}_p \cong \mathbb{Z}/p^{2k}\mathbb{Z}$$

with pairing

$$\langle x, y \rangle = \frac{xy}{p^{2k}} \in \mathbb{Q}/\mathbb{Z}.$$

The integral lattice $\mathbb{Z}_p/p^k\mathbb{Z}_p$ corresponds to $p^k\mathbb{Z}/p^{2k}\mathbb{Z}$ and is self-annihilating. Finite products of these local windows remain self-dual.

In conductor-normalized coordinates, the Loeb-Rosser boundary still dualizes by signed transpose:

$$D(d_S) = -d_S^T, \quad D^2(d_S) = d_S.$$

The obstruction appears at the diagonal quotient. For each finite conductor $N = \prod p^{e_p}$, CRT gives

$$\mathbb{Z}/N\mathbb{Z} \cong \prod_{p|N} \mathbb{Z}/p^{e_p}\mathbb{Z}.$$

So the finite-level quotient by the diagonal is zero. The phantom $\widehat{\mathbb{Z}}/\mathbb{Z}$ is therefore a limiting derived/pro phenomenon, not an ordinary finite-stage quotient.

Verification

The checker `code/scripts/check-pass67.py` generated `artifacts/reports/pass67-restricted-product-adelic-duality-check.json`.

Verified facts:

- local conductor quotients for $p = 2, 3, 5, 7$ have nondegenerate pairings;
- the integral lattice is self-annihilating in each tested local window;
- finite product windows preserve the self-annihilating lattice;
- signed boundary transpose checks pass for $s = 1, \dots, 7$;
- CRT diagonal maps are surjective for $N = 6, 12, 90, 420$.

Remaining Gap

Pass 68 should choose the exact category for the quotient: pro-abelian derived cokernel, condensed/solid quotient, or an exact-category extension class recovering the Loeb-Rosser phantom from levelwise-zero finite quotients.